

Executive Summary

Wheat (*Triticum* spp.) is the world's second most-produced cereal, with an estimated **799 million tonnes** in 2023 [14†L179-L187]. It is a staple for bread, pasta and many foods globally. Major producers include China, India, Russia, the United States, and France [14†L179-L187]. India alone produced ~98.5 Mt (2023) on ~30.6 Mha [42†L493-L500], with Uttar Pradesh (35.3 Mt), Punjab (17.7 Mt) and Haryana (11.2 Mt) accounting for the bulk of national output [14†L194-L202]. Wheat is grown from temperate zones into irrigated tropical highlands, favoring cool, moist seasons (daytime ~15–20 °C) but requiring a warm, dry ripening period [66†L98-L105]. Its grains are rich in carbohydrates (~72 g/100 g), with moderate protein (~13 g) and fiber (~11 g) [53†L239-L247]. Wheat soils are typically loams (pH 6–8) with good fertility; fertilizer needs can reach ~150 kg N, 40 kg P and 35 kg K per hectare for high yields [66†L110-L114]. Key production constraints include seasonal water deficits, heat stress, weeds, pests (aphids, borers, mites), and diseases (rusts, blights, smuts) that can cause 15–20% yield losses [24†L42-L49] [30†L129-L136]. Climate change poses further risk: models project global wheat yield declines (~–1.9% by mid-century) with severe losses (~15–16%) in vulnerable regions [59†L53-L61]. Adaptations – improved varieties (heat/drought tolerance), optimized planting dates, and integrated pest management – are essential to close yield gaps (now 5–25% in leading countries) [42†L493-L500] [59†L53-L61]. This report reviews wheat's biology and agronomy, varieties, production statistics, and challenges, with data and recommendations for management and future improvements.

Scientific Classification

Wheat belongs to the grass family Poaceae. Bread wheat (*Triticum aestivum* L.) is the common hexaploid ($2n=6x=42$, genome AABBDD) cultivated worldwide [7†L122-L130]. Durum wheat (*T. durum*, tetraploid AABB) is used for pasta and semolina. Other species include emmer (*T. dicoccum*), einkorn (*T. monococcum*), spelt (*T. spelta*), and triticale (wheat-rye hybrid). The genus *Triticum* is closely related to rye (*Secale*) and barley. A standard classification places *T. aestivum* under Poaceae, subfamily Pooideae [7†L122-L130].

Origin and History

Wheat was first domesticated ~10,000 years ago in the Fertile Crescent (Middle East) from wild einkorn and emmer ancestors [1†L110-L118]. It spread into Europe, Asia and North Africa by 5000 BC. Ancient varieties include emmer and einkorn, while modern bread wheat evolved via hybridization events to include three genomes [7†L122-L130]. Selective breeding accelerated in the 20th century: Norman Borlaug's semi-dwarf high-yield wheats in the 1960s sparked a "Green Revolution" in Asia. In India, wheat production rose from ~10.4 Mt (1965) to ~98 Mt (2017) thanks to improved cultivars and input technologies [36†L186-L194]. Today hundreds of improved varieties (e.g. HD2967, PBW342) are grown worldwide; India alone has **448 notified varieties** (378 bread, 59 durum, 7 emmer, 4 triticale) [36†L249-L254].

Botanical Description and Morphology

Wheat is an erect annual grass, 0.6–1.2 m tall. The culm (stem) has swollen nodes, and leaves are alternate, linear with parallel veins. Roots form a fibrous system. The inflorescence is a spike with sessile spikelets on a central rachis. Each spikelet (2–5 cm long) contains 2–8 florets enclosed by lemma and palea. **Grains** (caryopses) are small, single-seeded, oval to oblong, protected by glumes. Kernels have a tough bran, starchy endosperm, and small embryo. Wheat's **flowering** (anthesis) produces anthers which dehisce to release pollen [26†L123-L131] . Typical chromosome number is 42 (bread wheat) or 28 (durum).

Growth Stages and Phenology

Wheat development is described by the Zadoks scale, a decimal code from 0 (germination) to 9 (ripening) [26†L39-L48] [26†L69-L77] . Key stages: germination (Z0), tillering (Z2), stem elongation (Z3–4, “jointing” or booting), heading (Z5), flowering/anthesis (Z6), grain filling (Z7–8), and ripening (Z9). A winter wheat crop (~180–250 days) overwinters in juvenile stages and requires vernalization; spring wheat (100–130 days) is sown in spring and is day-neutral [66†L86-L94] [66†L98-L105] . Internally, flowering initiation coincides with double-ridge and terminal-spikelet development. The timing of each stage is influenced by temperature and photoperiod [26†L121-L130] [66†L98-L105] .

Climate and Soil Requirements

Wheat is adapted to cool, moist temperate climates, but also grown in subtropical irrigated zones. Optimal vegetative growth occurs at mean temperatures of ~15–20 °C [66†L98-L105] . The minimum threshold for growth is ~5 °C; frost tolerance is high in vegetative stages (down to –20 °C for winter types) but flowering and grain-fill are frost-sensitive [66†L98-L105] . A warm, dry ripening period (~18–22 °C) is preferred. Daylength affects vernalization and flowering. Soils of medium texture (loams, silts) are ideal. Optimal pH is 6.0–8.0 [66†L110-L114] . Soils with excessive soluble salts (EC >6 dS/m) or toxic levels of Na/Mg/Fe in peat are unsuitable. Wheat is moderately tolerant of shallow water tables (up to ~0.6 m) but yields drop if flooded [66†L115-L123] .

Land Preparation

Wheat requires a firm, fine seedbed to ensure proper seed-soil contact. Fields are typically ploughed or disked followed by leveling. In irrigated areas, puddling is avoided to prevent hard pan. Good drainage is essential to avoid waterlogging. Residue of previous crops is often chopped or incorporated.

Seed Selection, Rate, and Sowing Methods

Use certified, disease-free seed of adapted varieties. **Sowing depth** is shallow when moisture is good (2–4 cm) but deeper in dry seedbeds (5–8 cm) [66†L126-L133] . Typical **planting rates** are ~100–120 kg/ha for drilled wheat, or 110–140 kg/ha if broadcasted [66†L126-L133] . Row spacing is 12–15 cm under assured moisture (yield 0.45–0.70 million plants/ha), widened to ~25 cm under scarce moisture [66†L126-L133] . Sowing is done in the season suited to variety (November–December for north Indian plains; earlier in warmer regions). Precision sowing ensures even stand establishment.

Irrigation and Water Management

Wheat typically requires **450–650 mm** of water from sowing to maturity. In semi-arid regions, ~3–5 irrigations are applied. Crop coefficients (Kc) range from ~0.3 at germination to ~1.15 at anthesis

【66†L264-L272】. First irrigation is often at CRI stage (crown root initiation, Zadoks 30), followed by irrigation at tillering and booting if rain is lacking. A light pre-sowing irrigation may be used if soil moisture is low. Overirrigation is avoided to prevent lodging and disease. Efficient irrigation scheduling (e.g. based on soil moisture) can save water.

Nutrient and Fertilizer Management

For **high yields**, nutrients must be balanced. Soil tests determine specific needs. Typical fertilizer applications are up to 120–150 kg N/ha, 35–45 kg P₂O₅/ha and 25–50 kg K₂O/ha 【66†L110-L114】. Nitrogen is often applied in split doses (half basal, rest at tillering and booting). Phosphorus and potash are usually applied at sowing. On deficient soils, Zn and S may also be needed. Crop rotations (with legumes) and organic amendments (FYM) improve fertility. Integrated nutrient management optimizes yield while protecting the environment.

Weed Management

Weeds compete strongly with wheat, especially in early growth. Common **weeds** include grasses (e.g. *Phalaris minor* (little seed canarygrass), *Avena ludoviciana* (wild oat), *Lolium temulentum* (darnel), *Poa annua*), broadleaves (*Chenopodium album* (bathua), *Anagallis arvensis*, *Melilotus alba*, *Fumaria parviflora*) and sedges (*Cyperus rotundus*) 【30†L208-L212】. Heavy weed infestation can reduce yields by **15–30%** 【30†L129-L136】. Management includes clean tillage, timely planting, and raising seed rate (125–150 kg/ha) for competitive stands 【30†L159-L168】. One hoeing or mechanical weeding at 30–45 days after sowing is effective. Chemical control (selective herbicides) is used when weeds are problem; for example, pre-emergence herbicides (e.g. pendimethalin) and post-emergence options (e.g. 2,4-D, metsulfuron) target major weeds. Integrated approaches combine crop rotation (rice–wheat disrupts some grasses) and cultural methods 【30†L169-L178】.

Insect Pests

Major insect pests include:

- **Wheat aphids** (e.g. *Schizaphis graminum*, *Sitobion avenae*): feed on sap, transmit viruses; heavy infestations can cause yellowing and yield loss 【16†L290-L299】.
- **Pink (orange) stem borer** (*Sesamia inferens*): larvae bore stems and heads, causing dead hearts or white ears, especially in rice-wheat rotations 【16†L290-L299】.
- **Wheat midge** (*Sitodiplosis mosellana*): if present, larvae feed in developing grains.
- **Armyworms** (e.g. *Spodoptera litura*) and **army cutworm**: chewing defoliators.
- **Termites** (yellow termite, *Microtermes*, *Odontotermes*): attack roots/stems in warm areas.
- **Cutworms** (e.g. *Agrotis ipsilon*): cut seedlings at ground level.
- **Aphids** (e.g. *Rhopalosiphum maidis*, *Aphis craccivora*): in some regions.

These pests have multiple generations; population peaks often in early vegetative or heading stages. Detection via field scouting is essential.

Diseases

Important diseases include:

- **Rusts:**
 - *Stripe (yellow) rust* (*Puccinia striiformis*): bright yellow pustules in stripes on leaves; yields can drop sharply in cool, moist conditions [24†L42-L49] .
 - *Leaf (brown) rust* (*P. triticina*): small round brown pustules on leaf surfaces.
 - *Stem (black) rust* (*P. graminis*): black pustules on stems/leaf sheaths; historically devastating (e.g. Ug99 strain).
 - **Powdery mildew** (*Blumeria graminis*): whitish powder on leaves, thrives in humid weather.
 - **Blotch diseases:** *Tan spot* (*Pyrenophora tritici-repentis*) and *spot blotch* (*Bipolaris sorokiniana*): cause brown lesions on leaves [24†L42-L49] .
 - **Fusarium head blight (scab)** (*F. graminearum*): infections on spikes in warm, humid weather; causes bleaching, shriveled grain, mycotoxins.
 - **Karnal bunt** (*Tilletia indica*): grain-borne, causes partial bunt; important quarantine disease (limited distribution).
 - **Loose smut** (*Ustilago tritici*): replaces kernels with black spore masses.
 - **Flag smut** (*Urocystis agropyri*): black streaks on leaves and spores in grains.
 - **Take-all** (*Gaeumannomyces graminis*): root rot disease in cool, wet soils.
- These fungal diseases can reduce yields ~15–20% annually [24†L42-L49] . Symptoms (e.g. rust pustules, leaf spots, head blight) are monitored to time control measures.

Disease Management (IPM)

Integrated disease management is critical. Strategies include: using **resistant varieties**, seed treatment (fungicidal), crop rotation (to break pathogen cycles), balanced fertilization, and judicious fungicide use. For example, seed dressing with Triazoles/Benzimidazoles controls loose smut and bunt. Foliar fungicides (azoxystrobin, propiconazole) can manage rusts and foliar diseases. Timely fungicide sprays at flag leaf or flowering (for head blight) are often used. Biological control (antagonistic microbes) and cultural practices (deep plowing of stubble, removing volunteer plants) reduce inoculum. Overall, IPM combines cultural, biological, and chemical methods to minimize losses [16†L278-L284] .

Harvesting, Threshing, Drying, Storage and Losses

Wheat is harvested when grains reach ~20–25% moisture (typically in spring after maturity). Harvest timing should avoid shattering; swathings may be used in adverse conditions. Threshing (separating grain) is done via combines or stationary threshers. The grain is then dried to ~12–14% moisture for safe storage. Improper drying/storage can lead to spoilage: fungal growth (aflatoxins) or insect pests (weevils). In traditional storage (bags or bins), losses of 5–10% are common if not controlled. Proper storage (metal silos, chemical fumigation with phosphine or controlled atmosphere) reduces post-harvest losses. Careful handling, cleaning, and storage hygiene are recommended to maintain grain quality.

Processing and Value-Added Products

Wheat grains are primarily milled into flour (whole or refined) and semolina (from durum). **Major products** include:

- **Flour** for bread, chapatis (atta), biscuits, pasta (semolina from durum), noodles, cakes, etc.

- **Bran:** a by-product, used for animal feed or fiber-enriched foods.
 - **Wheat germ:** extracted oil, nutritional uses.
 - **Wheat straw:** used as fodder, bedding, or bioenergy feedstock.
 - Wheat is also fermented into **beer (malting)** and **bioethanol**, contributing to ~4 billion USD global wheat starch market [34†L217-L224] .
- Beyond food, wheat derivatives enter non-food industries (adhesives, biomaterials). In developed markets, product specification (protein content, gluten quality) adds value (premium grades for baking). By-products from straw and stillage (after ethanol) are used for livestock feed and organic acids [34†L217-L224] . These value-addition chains improve overall wheat sector economics.

Major Wheat Types and Varieties

Wheat falls into several types:

- **Bread wheat (T. aestivum)** – most common, used for leavened bread.
- **Durum wheat (T. durum)** – hard, high-protein wheat for pasta/semolina.
- **Emmer (T. dicoccum), einkorn (T. monococcum), spelt (T. spelta)** – ancient wheats grown on limited scales for niche markets (often with nutritional/heritage appeal).
- **Club wheats, Khorasan (Kamut)** – specialty wheats.

Numerous improved varieties have been bred. In India, for example, **448 varieties** are notified for cultivation (378 bread wheat, 59 durum, 7 emmer, 4 triticale) [36†L249-L254] , each adapted to specific zones or qualities (drought resistance, late-sowing, high protein). Globally, breeding efforts focus on yield, disease resistance, quality traits (gluten strength, micronutrient content), and abiotic stress tolerance. Examples: semi-dwarf lines like Mexican “Norin 10” derivatives in the 1960s; modern lines with rust resistance genes (e.g. Sr31, Yr18); recently, lines with genome edits for enhanced traits [61†L589-L597] [61†L598-L601] .

Comparison of Select Varieties (Examples)

Varietal Group	Use	Key Traits
Bread wheat (T. aestivum)	General bread baking	High yield, gluten strength
Durum wheat (T. durum)	Pasta, semolina	High protein, hard grains
Emmer/einkorn/spelt	Ancient wheats	Often higher micronutrients; niche markets
Improved Indian var. (e.g. HD2967, PBW343)	Indian plains	High yield, N-responsiveness
Drought/heat-tolerant lines	Marginal areas	Tolerance to water/heat stress

(Sources: ICAR-IIWBR variety catalogs [36†L249-L254])

Global and India Production Statistics, Trends, and Yield Gaps

Wheat is grown on ~220 Mha worldwide, yielding ~799 Mt in 2023 【14†L179-L187】. Top producers (2023) are (in descending order): China, India, Russia, USA, France, Canada, Pakistan, Australia, Ukraine and Germany 【14†L179-L187】. Per [42], 2017 production and average yield gaps (1960–2017) were:

Country	Production (2017)	Avg. Yield Gap (1960–2017)
China	134.0 Mt	12% 【42†L493-L500】
India	98.5 Mt	4% 【42†L493-L500】
Russia	85.9 Mt	21% 【42†L493-L500】
USA	47.4 Mt	12% 【42†L493-L500】
France	36.9 Mt	15% 【42†L493-L500】
Canada	31.8 Mt	24% 【42†L493-L500】
Australia	22.3 Mt	24% 【42†L493-L500】
Pakistan	26.7 Mt	8% 【42†L493-L500】
Turkey	21.5 Mt	12% 【42†L493-L500】
Germany	24.5 Mt	~0% 【42†L493-L500】

These data highlight that yield gaps remain large in many countries, indicating room for improvement. For instance, India's gap is relatively small (~4%) due to intensive irrigation and inputs, whereas Canada and Australia (rainfed) show ~24% gap 【42†L493-L500】. Global wheat area and production have generally increased over the past decades, driven by higher yields and expanded irrigation, though recent gains have slowed. (FAOSTAT reports: from ~600 Mt in 2000 to ~780 Mt in 2020).

State-wise in India, about 65% of production comes from northern states 【14†L194-L202】. **Top Indian states (2024):** Uttar Pradesh (35.34 Mt), Punjab (17.74 Mt), Haryana (11.19 Mt) 【14†L194-L202】; Madhya Pradesh, Rajasthan, Bihar, and Gujarat are other significant producers. For example, the state of Madhya Pradesh has been boosting production with early maturing varieties and irrigation.

Production Trends (Chart)

Figure: Wheat production has risen substantially since the 1960s in India (Green Revolution) and globally (FAO data). (Chart data not embedded here.) Yield gaps vary by region as shown above; addressing them is crucial to meet future demand.

Trade, Prices, and Economics

Wheat is one of the most traded grains globally (~180 Mt traded annually). The market is dominated by a few exporters: the European Union (EU as a bloc), Australia, Russia, Canada and Argentina are projected to account for ~80% of exports 【50†L31-L34】. Other major exporters include the United States and Ukraine. Top importers include countries in North Africa, the Middle East and Southeast Asia (e.g. Egypt, Turkey, Bangladesh, Indonesia).

International wheat prices spiked to record levels during 2022 (Russian–Ukraine conflict) and then eased in 2023 due to ample supply [51†L33-L40] . By 2023 prices were back near 2021 levels [51†L33-L40] . Futures markets (Chicago Board of Trade) typically range \$180–\$250/tonne; projected to stabilize around \$287/tonne by 2033 [51†L66-L72] .

On-farm economics vary by region. In irrigated northern India, yield potentials are high but input costs (seed, fertilizer, water) also rise, affecting profits. Minimum Support Prices (MSP) in India and subsidies can alter profitability. In developed countries, production costs (machinery, labor) are balanced by economies of scale. Across systems, profitability depends on yield level and input efficiency.

Nutritional Composition and Health Aspects

Composition: 100 g whole wheat flour (3.5 oz) contains ~340 kcal, 13.2 g protein, 72 g carbohydrates, 10.7 g fiber, 2.5 g fat [53†L239-L247] . It is rich in B-vitamins (especially niacin, folate), minerals (selenium, magnesium, phosphorus) and phytochemicals (phenolics in bran). Protein forms 7–14% of grain mass, mostly gluten (up to 80% of protein) [53†L297-L304] . Whole wheat has higher fiber (insoluble *arabinoxylans* ~70% of bran fiber) than refined flour [53†L272-L279] .

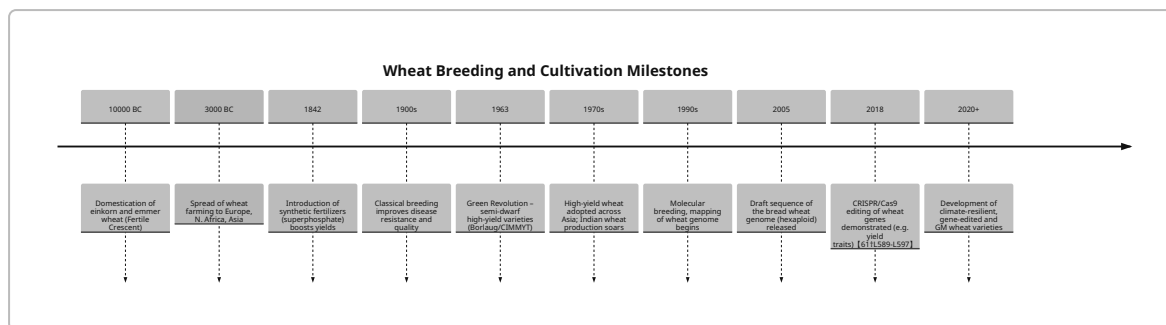
Health aspects: Whole-grain wheat aids digestion (fiber) and provides sustained energy. Dietary fiber lowers cholesterol and may benefit colon health. However, gluten (a wheat protein) triggers celiac disease (~1% of people) and non-celiac gluten sensitivity. Wheat also contains carbohydrates with moderate glycemic index, so diabetics moderate intake. Wheat bran/extracts are used for fiber supplements. Micronutrient content (e.g. selenium) depends on soil; some regions require fortification for full nutrition. Overall, when tolerated, wheat is a nutritious staple; healthwise it is valued for fiber and nutrients, but concerns exist for those with gluten intolerance.

Effects of Climate Change and Adaptation Strategies

Climate change threatens wheat yields through increased heat stress, drought, and erratic rainfall. Projections indicate global yields may decline modestly (~–1.9% by 2050) [59†L53-L61] , but impacts are severe in already warm regions (~–15% in Africa, ~–16% in Southern Asia) [59†L53-L61] . In India, average temperature rise (~2.2 °C by 2050) could cut yields by up to 20% [14†L253-L259] . Warmer winters may reduce vernalization (affecting flowering), while more extreme rains or heat waves at anthesis can sterilize spikes.

Adaptation strategies include:

- **Breeding:** Develop heat- and drought-tolerant varieties, early-maturing lines to escape terminal heat. Gene banks and biotechnology (marker-assisted selection, CRISPR) enable faster development [59†L53-L61] [61†L589-L597] . Examples: cultivars with deep roots, efficient water use, or heat-shock protein expression.
- **Agronomy:** Adjust sowing dates (earlier planting to avoid peak heat) and varieties to local climate. Conservation agriculture (mulching, residue retention) conserves soil moisture. Improved irrigation (e.g. drip) and deficit irrigation scheduling can offset drought.
- **Integrated soil–water management:** Conservation tillage, intercropping or cover crops improve resilience. Simulations show that combining genetic improvements (new traits) with optimized N fertilizer can mitigate climate impacts in high-input systems [59†L53-L61] . However, technology adoption and infrastructure are critical for adaptation. Overall, climate-smart practices and robust varieties are needed to sustain wheat productivity under change.



Modern Technologies in Wheat Agriculture

Recent advances are transforming wheat farming:

- **Precision Agriculture:** GPS-guided planting and harvesting, variable-rate fertilizer application based on field sensors, and yield monitors improve input use efficiency. Satellite/drone imagery detects nutrient or pest hotspots for targeted action.
- **Drones and Remote Sensing:** Unmanned aerial vehicles carry multispectral cameras to monitor crop health, stress, and nutrient status in real time, enabling site-specific interventions.
- **Data Analytics & AI:** Machine learning models use weather, soil and genetic data to forecast yields and identify stress patterns. Decision support apps help farmers optimize planting dates and inputs.
- **Biotechnology:** While **no commercial GM wheat** is yet grown (difficult polyploid genetics) [61†L598-L601], advanced tools are in use. CRISPR-Cas9 editing has successfully modified multiple wheat genes simultaneously, producing larger grains and disease resistance in trials [61†L589-L597]. Transgenic approaches (e.g. RNAi for virus resistance) are under research. Marker-assisted and genomic selection accelerates variety development.
- **Farm Machinery:** Modern combines, planters, and irrigation systems (e.g. pivots, drip) increase scale and precision of operations.

Challenges and Future Prospects

Wheat faces challenges of diminishing arable land, climate stress, and evolving pests/pathogens. Soil degradation, water scarcity, and trade disruptions are concerns. To meet rising food demand, wheat yield growth must accelerate. Closing yield gaps through improved genetics (high-yielding, stress-tolerant varieties), better agronomy (nutrient and water use efficiency), and resilience strategies is key. Global breeding efforts (public and private) are focusing on biofortification (e.g. higher iron, zinc) and stress tolerance. Precision agriculture and digital tools will enhance decision-making. Climate change adaptation will rely on both traditional breeding and novel approaches (e.g. gene editing). Despite challenges, wheat's critical role in food security ensures continued innovation and investment. Continued emphasis on integrated management (crop rotation, soil health, market support) will be needed to sustain production.

Sources: Authoritative references on wheat agronomy and statistics were used, including FAO and ICAR data, peer-reviewed studies and institutional reports [1†L110-L118] [14†L179-L187] [24†L42-L49] [30†L129-L136] [42†L493-L500] [53†L239-L247] [59†L53-L61] [61†L589-L597].